

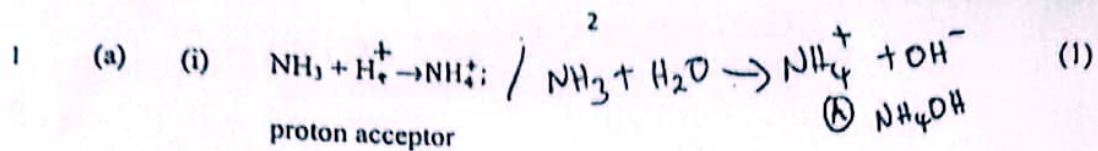
ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

MARKING SCHEME

NOVEMBER 2020 SESSION

CHEMISTRY

6031/2

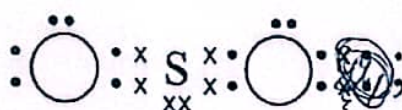


(ii) 1. Ammonia is less basic than methylamine; /A W (1)

CH_3 /alkyl group in methylamine has positive inductive effect/releases e^- to nitrogen atom making it more basic; (1)
 (pushes)

2. phenylamine is less basic than ammonia; (1)
 lone pair of electrons on nitrogen atom on phenylamine is delocalized; (1)

(b) (i)



one mark

(ii) bent/v'-shaped; non linear (1)
~~lone pair lone pair repulsion, on the oxygen atoms pushes the bond pair;~~ / converse VBEP (A) 2 bond pairs and 1 lone pair (1)
~~lone pair lone pair repulsion, on the oxygen atoms pushes the bond pair;~~

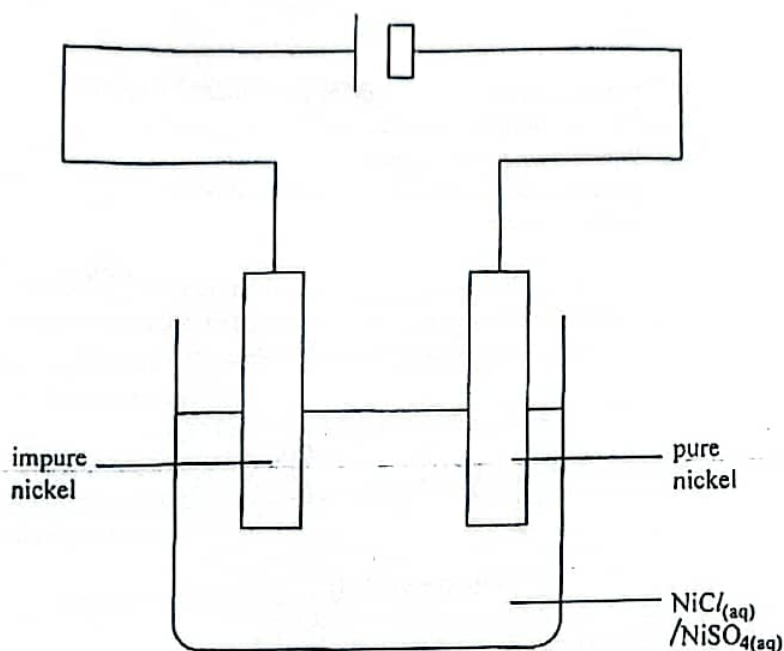
2 (a) (i) graphite (both electrodes); carbon (1) LP-BP > BP-BP (1)

(ii) Fluorine gas is toxic/
 toxic fumes of F_2/CO ;

(A) toxic gas

heat
 greenhouse (A W)
 electrocution

(b) (i)



One mark both electrodes correct; *diag even without label* (1)
 One mark correct electrolyte; *label* (1)

(ii) mass decreases/becomes thinner; *AW* (1)

(c) (i) $\left(\begin{array}{l} \text{Br}_2 + 2e^- \rightarrow 2\text{Br}^- \quad E^\theta + 1,07 \\ \text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6e^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O} \quad E^\theta + 1,33 \end{array} \right)$ (1)

$1,33 - 1,07 = +0,26$; *(no sign no mark)* (1)

(ii) $+6 \text{ to } +3$ / decreases by 3; *give all to candidates writing the correct answer* (1)

(d) E^θ_{cell} for $2\text{Ag} + 2\text{H}^+ \rightarrow \text{H}_2 + 2\text{Ag}^+ = -0,80$; *Eqn (1) cal (1)* (1)

reaction is not feasible; *ating 2* (1)

3 (a) (i) $\text{Ca}(\text{OH})_2 + 2\text{NH}_4\text{Cl} \rightarrow \text{CaCl}_2 + 2\text{H}_2\text{O} + 2\text{NH}_3$; *correct ion length state symbols* (1)

(ii) drying agent; /AW- absorb water (1)

(iii) it is neutralized with the ammonia; /it reacts with ammonia (in an acid-base reaction;)
 lower yield of ammonia; (1)

(b) (i) excess nitrates cause algal bloom/bloom of algae/AW
 when algae dies, bacteria act on it;
 bacteria respire aerobically/uses up oxygen;
 aquatic life which uses dissolves oxygen dies;

Or

excess nitrates cause algal bloom/bloom of algae;
blocks sunlight from reacting aquatic plants,
photosynthesis of aquatic plants,
causing less dissolved oxygen/aquatic dies;
(Any 3 points)

(30) first 3
also 100%

- (ii) adding/applying most economical amounts of fertilizer;
/apply fertilizer at the right time; /use of organic manure;
/use of nanofertilisers; *crop rotation any* (1)

- (c) (i) $E/\theta = V.c. \Delta\theta = 200 \times 4.18 \times 27 = 22572 \text{ J};$ (1)

$$\Delta H = \frac{22572 \text{ (J)}}{\frac{1.04}{32}} = \frac{32 \times 22572}{1.04}$$

(100%)

$$= -695 \text{ kJmol}^{-1}; (1)$$

or

$$-694.5 \text{ kJmol}^{-1}$$

- (ii) loss of heat (by convection/radiation/conduction;)
/in complete combustion; (1)

- 4 (a) (i) *std condition may not be met*
nitrogen has a lone pair of electrons; which enables it to accept a proton. (1)

- (ii) $\text{C}_6\text{H}_5\text{NH}_2 < \text{NH}_3 < (\text{CH}_3)_2\text{NH}$
least basic most basic

lone pair on nitrogen in $\text{C}_6\text{H}_5\text{NH}_2$ is delocalized into ring
less available for dative bonding to H^+ ; (1)

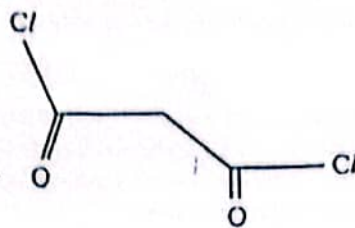
hydrogen has no effect on the lone pair of nitrogen in ammonia/AW (1)
the two methyl groups pushes electrons cloud.
towards nitrogen making lone pair more available; (1) (1)

- (b) (i) condensation/addition-elimination; (1)

- (ii) (1)



(1)



(1)

(iii) chains held together by weak vDw forces/no cross linkages between chains;

(1)

(c) HCN is a weak acid/ $\text{HCN} \rightleftharpoons \text{H}^+ + \text{CN}^-$;
acids shift the equilibrium to the left hence no cyanide (CN^-) to attack;
carboxyl carbon AW

(1)

(1)

5

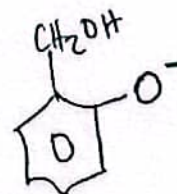
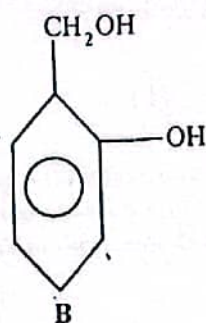
(a) (i) Reagent - CH_3COCl

(1)

Condition - reflux ^{or heat} in (alkaline solution) ^{ET & P ✓✓} (strong base) ^(OH⁻)

(1)

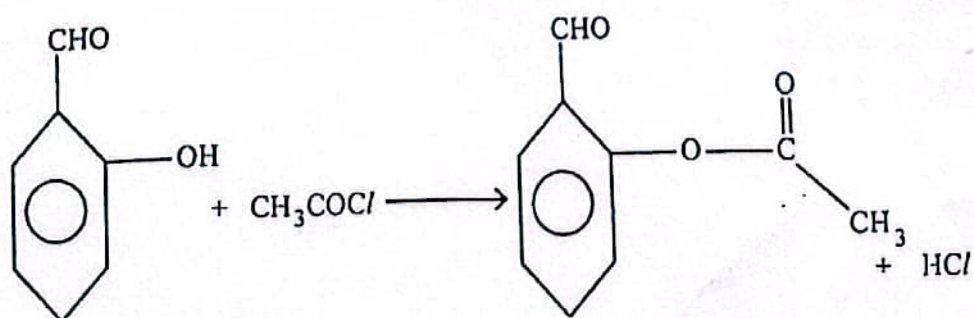
(ii)



(iii) electrophilic substitution

(1)

(iv)



(1)

(v) white fumes and ~~and~~ sweet smell;

(1)

- (b) (i) nitro group is electron withdrawing (deactivating) (1)
making the benzene ring less susceptible to electrophilic attack; (1)
- (ii) methylbenzene form van der waals interactions (1)
(hydrophobus) with CCl_4 hence it dissolves
methylbenzene cannot form hydrogen bonds with water thus
it does not dissolve in water (1)
- 6 (a) (i) artificial limbs; (1)
drugs delivery; (1)
cosmetics; ~~(1)~~
sunscreen; ~~(1)~~
nanosensors; ~~(1)~~
for diagnostics; ~~(1)~~
treatment of cancer *implant surgery production*
tissue regeneration *absorb forms, 3 drugs*
- (ii) toxic; (1)
bioaccumulation; (1)
enter the environment; *any two (consider any potential risk.)*
enter into explosion/fire
- (h) (i) high $[\text{H}^+]$; equilibrium shifts to the left; (2)
more H/n molecules producing colourless solution;
- (ii) $\frac{[\text{In}^-]}{[\text{In}]} = \frac{7.0 \times 10^{-10}}{10^{-9}}; (1)$
 $= 0.7 / \frac{7}{10}; (1)$
- (iii) Titration of a strong acid with weak base; ~~(1)~~
equivalent end point does not coincide with pH range over which
phenolphthalein changes colour/AW (1)